ZS-Q7: Structural Arrow of Time from the Z-Bottleneck: Entropy Production without a Past Hypothesis

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Verification: 33/33 PASS | Zero Free Parameters

§0. Abstract

We derive a structural arrow of time within Z-Spin cosmology from the single geometric impedance $A = 35/437$ with zero new parameters. The derivation proceeds in four stages: (1) unitary evolution on the full Hilbert space $H_X \otimes H_Z \otimes H_Y$, (2) Stinespring dilation yielding a CPTP channel (ZS-Q1 v1.0 §3.3, PROVEN), (3) Lindblad master equation with $\Gamma = 2A(\Delta E/\hbar)^2$ (ZS-Q1 v1.0 §3.4, DERIVED), and (4) sector coarse-graining to a Pauli master equation. The Born-Markov approximation is quantitatively justified by the validity parameter $\varepsilon_{BM} = \tau_{fast}/\tau_{slow} = 2/Q = 2/11 \ll 1$, a purely geometric ratio.

Three theorems are established: (I) the Dimension Ratio Theorem (PROVEN): $\Gamma(X\to Y)/\Gamma(Y\to X) = \dim(Y)/\dim(X) = 2$ is an exact mathematical identity from trace cyclicity; (II) the Z-Bottleneck Channel Bound (DERIVED): $\mathrm{rank}(T_{XY}) \le \dim(Z) = 2$ and channel capacity $\le \ln(2)$; (III) the Z-Mediated Master Equation (DERIVED): $D_{KL}(p \,\|\, p_{eq})$ is a monotonically decreasing Lyapunov function with Schnakenberg entropy production $\sigma \ge 0$ everywhere.

(IIIA) The Exact Eigenvalue Theorem (DERIVED): the characteristic polynomial factorizes as $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$, giving $\tau_{fast} = 1/A$ exactly—coinciding with the decoherence ratio $\tau_D/\tau_{Penrose}$ from ZS-Q1 v1.0. The physical time scale is $\tau_D = \hbar/(A\cdot E_{diff})$, providing concrete predictions: for gold nanospheres (10^9 amu), $\tau_{relax} \approx 474$ days.

The resulting Structural Arrow produces entropy $\Delta S = \ln(2)$ per Z-mediated transition without a Past Hypothesis. The non-uniform equilibrium $p_{eq} = (3, 2, 6)/11$ reflects gravitational entropy dominance—the Y-sector (dim = 6) carries twice the state density of the X-sector (dim = 3), geometrically realizing Penrose's intuition. The Cosmological Closure Mapping (§5.8) derives all 8 Z-Sim closure parameters from the master equation, eliminating all phenomenological freedom. Zero free parameters, 7 falsification gates, 5 non-claims, 33-test verification suite with 100% pass rate.

Epistemic Status Legend

STATUS	DEFINITION	EXAMPLE	FALSIFIABLE?
PROVEN	Mathematical theorem from (Z,X,Y)=(2,3,6)	Γ ratio = 2	Only by logic error
DERIVED	Conditional on Z-Spin action being correct	$dD_{KL}/dt \le 0$	Yes (experiment)
NON-CLAIM	Explicitly excluded from scope	NC-Q7.1–5	N/A

LOCKED	Input fixed from prior paper	A = 35/437	N/A
VERIFIED	Numerically confirmed to stated precision via verification suite	Eigenvalue factorization	By code failure
CONSISTENT	Cross-paper consistency check; not an independent derivation	Wald entropy	By contradiction
OPEN	Recognized gap requiring future work	Spectral gap	N/A
BLOCKING	Gate that would invalidate the paper if triggered	F-Q7.6	Yes (structural)
PASSING	Gate currently satisfied by verification suite	F-Q7.4, F-Q7.7	Yes (computational)

§1. Introduction

1.1 The Arrow of Time Problem

The fundamental laws of physics are time-reversal symmetric at the microscopic level, yet the macroscopic world exhibits a profound asymmetry: entropy increases, eggs break but do not unbreak. This tension between microscopic reversibility and macroscopic irreversibility is the arrow of time problem.

Three principal approaches exist. Boltzmann's H-theorem requires a low-entropy initial state but cannot explain why it was special. Penrose's Weyl Curvature Hypothesis invokes the Past Hypothesis as an additional postulate. Carroll's "Mad-Dog Everettian" approach embeds the arrow in quantum branching but involves untestable multiverse claims. All three require either special initial conditions or additional postulates beyond the dynamical laws.

1.2 The Z-Spin Approach: Structural Asymmetry

Z-Spin cosmology offers a fundamentally different resolution. The sector decomposition $Q = Z + X + Y = 2 + 3 + 6 = 11$ (PROVEN, ZS-F5 v1.0) with $L_{XY} \equiv 0$ (PROVEN, ZS-F1 v1.0) creates a structural asymmetry: the Y-sector (dim = 6) has twice the dimension of the X-sector (dim = 3), and all transitions must pass through the Z-sector (dim = 2) bottleneck. This is not an initial condition but a geometric property of the theory itself.

1.3 Scope and Relation to ZS-A6

ZS-A6 v1.0 §6 introduced the structural arrow as a corollary. This paper elevates it to a self-contained treatment with five new contributions: (1) a rigorous derivation chain from unitary dynamics to the master equation (§5.0), (2) proof that the transition rate ratio is a mathematical identity (Theorem 1), (3) the Z-Bottleneck Channel Bound (Theorem 2), (4) the complete Pauli master equation with KL divergence as the correct Lyapunov function (Theorem 3), and (5) exact analytical eigenvalues with physical time scale identification (Theorem 3A).

§2. Locked Inputs

All quantities are locked from prior papers. Zero new parameters are introduced.

Quantity	Value	Source	Status
A (geometric impedance)	35/437 = 0.080092	ZS-F2 v1.0	LOCKED
(Z, X, Y) dimensions	(2, 3, 6); Q = 11	ZS-F5 v1.0	PROVEN
L_{XY}	≡ 0 (exact)	ZS-F1 v1.0	PROVEN
κ (coupling)	$\sqrt{(A/Q)}$ = 0.08533	ZS-S1 v1.0	DERIVED
τ_D (decoherence time)	$\hbar / (A \cdot E_{diff})$	ZS-Q1 v1.0 §5.1	DERIVED
Γ (Lindblad rate)	$2A(\Delta E/\hbar)^2$	ZS-Q1 v1.0 §3.4	DERIVED

§3. Theorem 1: Dimension Ratio Identity [PROVEN]

3.1 Statement and Proof

Theorem 1 (Dimension Ratio Identity). For any linear map $T: H_A \to H_B$ with $\dim(H_A) = d_A$, $\dim(H_B) = d_B$, the uniform-averaged transition rate ratio is:

$$\Gamma(A\to B) / \Gamma(B\to A) = d_B / d_A \qquad (1)$$

Proof. By trace cyclicity, $\mathrm{Tr}(T^\dagger T) = \Sigma_k \sigma^2_k = \mathrm{Tr}(TT^\dagger)$. Therefore $\Gamma(A\to B)/\Gamma(B\to A) = [\mathrm{Tr}(T^\dagger T)/d_A] / [\mathrm{Tr}(TT^\dagger)/d_B] = d_B/d_A$. □

[STATUS: PROVEN] Pure mathematical identity. No physics assumptions required. Variance = 0 across 10^4 random matrices.

3.2 Key Insight

Theorem 1 holds for any linear map T between any two Hilbert spaces. The ratio 2 is a mathematical certainty, not a statistical averaging result. Z-Spin's contribution is not the ratio itself, but the mechanism ($L_{XY} = 0$) that makes this ratio the only physical pathway.

§4. Theorem 2: Z-Bottleneck Channel Bound [DERIVED]

Theorem 2. If $L_{XY} \equiv 0$, then $T_{XY} = V_{ZY} \cdot V_{XZ}$, so $\mathrm{rank}(T_{XY}) \le \dim(Z) = 2$ and channel capacity $\le \ln(2)$.

Proof. $L_{XY} = 0$ prohibits first-order X→Y propagation. Heat kernel: $\|K_{XY}(t)\| \sim t^2$ (ZS-M6 v1.0 §4.5). All paths factor X→Z→Y, forcing $\mathrm{rank}(T_{XY}) \le \dim(Z) = 2$. □

[STATUS: DERIVED] From $L_{XY} = 0$ (PROVEN) + Block Laplacian structure.

Physical consequences: (i) information channel capacity $\leq \ln(2) \approx 1$ bit; (ii) shortcut prohibition ($L_{XY} = 0$ blocks bypass); (iii) irreversible information loss (3-dim info through 2-dim bottleneck). Theorem 1 determines the magnitude (ratio = 2); Theorem 2 provides the mechanism (Z-bottleneck) that forces this asymmetry to be physically realized.

§5. Theorem 3: Z-Mediated Master Equation [DERIVED]

§5.0 From Unitary Evolution to Master Equation

A critical question arises: the full universe evolves unitarily, yet we employ an irreversible master equation. This section demonstrates that the master equation is not an assumption but a derived consequence of unitary dynamics plus coarse-graining. The derivation proceeds in four steps:

Step 0 (Full Hilbert space): $H_{total} = H_X \otimes H_Z \otimes H_Y$ (dim = 36). $H = H_X + H_Z + H_Y + V_{XZ} + V_{ZY}$, with $V_{XY} = 0$ from $L_{XY} \equiv 0$.

Step 1 (Stinespring → CPTP): $\rho_X(t) = \mathrm{Tr}_{\{Z,Y\}}[U(t)\, \rho(0)\, U^\dagger(t)]$. By ZS-Q1 v1.0 §3.3, this yields a CPTP channel with dim(Z) = 2 Kraus operators.

Step 2 (SSE → Lindblad): $F(\varepsilon) = 1 + A\varepsilon^2$ generates stochastic phase kicks. $d\rho/dt = -(i/\hbar)[H, \rho] + \Gamma(\sigma_z \rho \sigma_z - \rho)$ with $\Gamma = 2A(\Delta E/\hbar)^2$.

Step 3 (Sector coarse-graining → Pauli master equation): $dp/dt = M \cdot p$ with $W_{AB} = \dim(B) \times A/Q$ from Fermi's golden rule.

5.0.1 Born-Markov Approximation: Quantitative Justification

$$\varepsilon_{BM} = \tau_{fast} / \tau_{slow} = 2/Q = 2/11 \approx 0.18 \qquad (4)$$

This ratio is purely geometric—it depends only on Q = 11, not on A or any dynamical parameter. For $Q \to \infty$, $\varepsilon_{BM} \to 0$ and the Markov approximation becomes exact.

5.1 Construction

$$dp_X/dt = -W_{XZ}\, p_X + W_{ZX}\, p_Z \qquad (5a)$$

$$dp_Z/dt = W_{XZ}\, p_X - (W_{ZX} + W_{ZY})\, p_Z + W_{YZ}\, p_Y \qquad (5b)$$

$$dp_Y/dt = W_{ZY}\, p_Z - W_{YZ}\, p_Y \qquad (5c)$$

with rates $W_{AB} = \dim(B) \times A/Q$. The ratio $W_{ZY}/W_{ZX} = \dim(Y)/\dim(X) = 2$ is Theorem 1 realized within the master equation.

5.2 Equilibrium Distribution

$$\mathbf{p_eq = (dim_X, dim_Z, dim_Y)/Q = (3, 2, 6)/11} \qquad \mathbf{(6)}$$

This is not the uniform distribution (1/3, 1/3, 1/3). The Y-sector (gravity, dim = 6) carries 54.5% of the probability, reflecting the dominance of gravitational degrees of freedom.

5.3 KL Divergence as Lyapunov Function

Theorem 3 (KL Monotonicity). D_KL(p(t) || p_eq) is a monotonically decreasing function of time:

$$dD_KL/dt \leq 0, \quad \text{equality iff } p = p_eq \qquad (7)$$

[STATUS: DERIVED] From Fermi golden rule + L_XY = 0 + microscopic reversibility.

5.4 Shannon Entropy Is NOT the Correct Lyapunov Function

Because p_eq = (3, 2, 6)/11 is non-uniform, Shannon entropy $S(p) = -\Sigma\, p_i \ln p_i$ is not monotonically increasing. Starting from p = (1, 0, 0), Shannon entropy rises to $S \approx 1.040$ (overshooting $S_eq \approx 0.995$) before decreasing. The correct potential is D_KL, which monotonically decreases. The Schnakenberg entropy production $\sigma = \frac{1}{2} \Sigma\, (J_\alpha\beta - J_\beta\alpha) \ln(J_\alpha\beta/J_\beta\alpha) \geq 0$ provides a complementary guarantee.

5.5 Theorem 3A: Exact Eigenvalue Factorization

Theorem 3A (Exact Eigenvalues). The characteristic polynomial of the transition matrix M factorizes as:

$$\boldsymbol{\lambda(\lambda + 2A/Q)(\lambda + A) = 0} \qquad \mathbf{(8)}$$

with eigenvalues $\lambda_0 = 0$, $\lambda_1 = -2A/Q$, $\lambda_2 = -A$ and relaxation timescales:

$$\tau_0 = \infty \text{ (stationary)}, \quad \tau_slow = Q/(2A), \quad \tau_fast = 1/A \qquad (9)$$

Physical interpretation: (i) τ_fast = 1/A = 437/35 = 12.4857... coincides exactly with τ_D/τ_Penrose from ZS-Q1 v1.0. (ii) $\lambda_1 = -2A/Q$ governs inter-sector thermalization, slower by Q/2 = 5.5. (iii) ε_BM = $|\lambda_1|/|\lambda_2|$ = 2/Q = 2/11 is purely geometric.

5.6 Physical Time Scale

$$\tau_D = \hbar / (A \cdot E_diff) \qquad (10)$$

Concrete predictions for gold nanospheres (10^9 amu, R ≈ 50 nm):

Quantity	Value	Timescale
τ_Penrose	$\hbar$/E_diff	≈13.3 hours
τ_D (Z-Spin decoherence)	τ_Pen / A	≈6.9 days
τ_fast (bottleneck relaxation)	τ_D / A = τ_Pen / A²	≈86 days
τ_slow (thermalization)	(Q/2) × τ_fast	≈474 days

5.7 Entropy Production Rate

$$dS_Z/dt \propto A \times (dim_X/Q) \times \ln(dim_Y/dim_X) = (35/437) \times (3/11) \times \ln(2) \approx 0.01514 \text{ nats/transition} \quad (12)$$

5.8 Cosmological Closure Mapping

The Pauli master equation with W_AB = dim(B)·A/Q provides a complete, parameter-free specification of cosmological sector energy exchange. Matching coefficients: γ_xz = 2A/Q = 0.01456, α_xz = dim(X)/dim(Z) = 3/2, γ_zy = 6A/Q = 0.04369, α_zy = dim(Z)/dim(Y) = 1/3. The ratio γ_zy/γ_xz = 3 = dim(Y)/dim(Z) is structural. Equilibrium initial conditions: p_eq = (3, 2, 6)/11. All 8 Z-Sim closure parameters are now DERIVED. [STATUS: DERIVED]

§6. Corollary: Structural Arrow of Time [DERIVED]

6.1 Synthesis

Combining Theorems 1–3A: (i) D_KL monotonically decreases from any initial state (Thm 3). (ii) Net probability current flows X→Y through Z with rate ratio 2:1 (Thm 1). (iii) No shortcut bypasses the Z-bottleneck (Thm 2). (iv) Each Z-mediated transition produces $\Delta S = \ln(2)$. (v) Relaxation occurs on timescale τ_D with exact eigenvalues $\lambda = 0, -2A/Q, -A$ (Thm 3A).

6.2 Past Hypothesis Unnecessary

Whether the universe began in a low-entropy state, high-entropy state, or random state, the asymmetric relaxation toward p_eq occurs identically. The arrow is not a consequence of where the universe started but of how the state space is structured.

6.3 Rapidity Asymmetry

The curvature rapidities ψ_X = artanh(5/19) = 0.2695 and ψ_Y = artanh(7/23) = 0.3143 satisfy $\Delta\psi$ = 0.0448 > 0 [PROVEN]. The Y-sector carries more curvature “momentum” than X.

6.4 Gravitational Entropy Dominance

The non-uniform equilibrium p_eq = (3, 2, 6)/11 reflects a fundamental asymmetry: the Y-sector (gravitational modes, dim = 6) dominates the entropy budget by factor 2 over the X-sector (matter modes,

dim = 3). This is the Z-Spin geometric realization of Penrose's physical intuition—derived from polyhedral geometry rather than assumed as an initial condition.

§7. Comparison with Existing Approaches

Criterion	Boltzmann	Penrose	Carroll	Z-Spin (Q7)
Initial conditions?	Required	Required	No	No
Past Hypothesis?	Yes	Yes (Weyl)	No	No
Free parameters	S_initial	0 (postulate)	0	0
Testable?	Indirect	Indirect	No	Yes (7 gates)
Entropy quantum	N/A	N/A	N/A	ln(2) per step
Time scale derived?	No	No	No	Yes ($\tau_D = \hbar/AE$)
Markov justified?	Assumed	N/A	N/A	Yes ($\varepsilon=2/Q$)

§8. Quantum Information Perspective

The Z-mediated transition is a CPTP channel Φ_Z: $B(H_X) \to B(H_Y)$ via Stinespring dilation (ZS-Q1 v1.0): $\Phi_Z(\rho_X) = \mathrm{Tr}_Z[V(\rho_X \otimes |0\rangle\langle 0|_Z)V^\dagger]$, with V: $H_X \otimes H_Z \to H_Y \otimes H_Z$ isometry. Properties: (a) output rank ≤ 2, (b) Holevo capacity $\chi \leq 1$ bit, (c) structural entropy production.

§9. Verification Suite

33 tests organized in 7 categories. All PASS.

Category	Tests	Result	Key Test
A. Locked Constants	5	5/5	A=35/437, Q=11, L_XY=0
B. Theorem 1 (Dim Ratio)	5	5/5	10^4 random matrices, var=0
C. Theorem 2 (Z-Bottleneck)	5	5/5	10^4 trials, max rank=2
D. Theorem 3 (Master Eq.)	5	5/5	D_{KL} monotone, $\sigma\geq 0$
E. Theorem 3A (Eigenvalues)	3	3/3	$\lambda_2=-A$ exact, $\varepsilon_{BM}=2/Q$
F. Anti-Numerology	5	5/5	Ratio tracks dim, not A
G. Cross-Paper Consistency	5	5/5	ZS-Q1 τ_D, ZS-A6 §6 compat

§10. Falsification Gates

Gate	Falsification Condition	Type	Status
F-Q7.1	CPT-independent T-violation observed with magnitude independent of A	Experimental	OPEN
F-Q7.2	Isolated single-body shows spontaneous irreversibility without Z-mediation	Experimental	OPEN

F-Q7.3	L_XY ≠ 0 discovered in Z-Spin action	Theoretical	OPEN
F-Q7.4	dD_KL/dt > 0 under Z-Spin compatible dynamics	Computational	PASSING
F-Q7.5	Z-Spin hardware (3,2,6) transition statistics ≠ dimension ratio	Experimental	OPEN (2028+)
F-Q7.6	dim(X) = dim(Y) in physical sector decomposition	Structural	BLOCKING
F-Q7.7	Master eqn eigenvalue $\lambda_2 \neq -A$	Computational	PASSING

§11. Non-Claims

NC-Q7.1: This arrow is a coarse-grained entropy bias. No CPT violation is claimed. Individual transitions are bidirectional.

NC-Q7.2: ΔS = ln(2) and γ_LQG = ln(2)/(π√3) may be structurally related, but this paper does NOT derive the Immirzi parameter.

NC-Q7.3: This paper derives the thermodynamic arrow. The cosmological arrow (expansion direction) is ZS-U scope.

NC-Q7.4: The arrow does not explain conscious temporal experience.

NC-Q7.5: The Z-bottleneck eliminates the Past Hypothesis but does NOT quantitatively suppress Boltzmann brain rates.

§12. Conclusion

We have established a structural arrow of time within Z-Spin cosmology through three theorems plus a subsidiary eigenvalue theorem, with zero new parameters. The master equation is derived from unitary dynamics via Stinespring → CPTP → Lindblad → Pauli master equation. The Born-Markov approximation is justified by ε_BM = 2/Q = 2/11. The exact eigenvalue factorization λ(λ + 2A/Q)(λ + A) = 0 gives τ_fast = 1/A exactly, coinciding with τ_D/τ_Penrose. The Cosmological Closure Mapping derives all 8 Z-Sim closure parameters. Shannon entropy is not the correct Lyapunov function. The Y-sector's dominance realizes Penrose's gravitational entropy intuition from geometry. Seven falsification gates and five non-claims. Priority: F-Q7.5 (Z-Spin hardware, ~2028–2032).

Appendix A. Cross-Reference Table

Paper	Input to ZS-Q7	Status	Section
ZS-F1 v1.0	Action S, L_XY = 0	PROVEN	§2, §4, §5.0
ZS-F2 v1.0	A = 35/437, δ_X, δ_Y	PROVEN	§2
ZS-F5 v1.0	(Z,X,Y)=(2,3,6), Q=11	PROVEN	§2, §3
ZS-Q1 v1.0 §3.3	Stinespring → CPTP	PROVEN	§5.0 Step 1
ZS-Q1 v1.0 §3.4	SSE → Lindblad, $\Gamma=2A(\Delta E/\hbar)^2$	DERIVED	§5.0 Step 2

ZS-Q1 v1.0 §5.1	$\tau_D = \hbar/(A \cdot E_diff)$	DERIVED	§5.5, §5.6
ZS-M6 v1.0	$K_XY \sim t^2$ (two-step)	VERIFIED	§4
ZS-A6 v1.0 §6	Arrow (precursor)	DERIVED	§1, §6
ZS-T3 v1.0	Z-Sim forward simulator	CONSISTENT	Cross-check

Acknowledgements & Code Availability

Acknowledgements. This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (ZS-Q7_v1_0_verification.py) is publicly available. Dependencies: Python 3.10+, NumPy, SciPy, fractions (stdlib). Execution: python3 ZS-Q7_v1_0_verification.py. Expected output: 33/33 PASS, exit code 0.

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v1.2.0)